Ardhya Novri Ramadhan

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Bachelor of Engineering in Electrical Engineering from Diponegoro University, with experience in electronics, airport electrical systems, student organizations and community projects. Develops embedded monitoring systems and AI-assisted software, and is the sole developer of ENVIID's company website.

EDUCATION

Diponegoro University Aug 2022 - Sep 2026

Bachelor of Engineering in Electrical Engineering

GPA: 3.43/4.00. Graduated on 29 September 2026.

SMA Negeri 5 Bekasi Jul 2019 - May 2022

Natural Sciences

PROFESSIONAL AND COMMUNITY EXPERIENCE

ENVIID Carbon Solutions Oct 2026 - Present

Chief Technology Officer / IT & Digital Lead

Sole developer of the company website, built with AI assistance, including project information, a carbon calculator and a digital MRV workflow demonstration.

Diponegoro University Jan 2026 - Feb 2026

Community Service Participant (KKN) Mutihan, Klaten

- Helped assemble and install a solar-powered Smart Owl House, including sensors, battery and controller. Prepared operating and maintenance guidance.
- Assisted village administration, Posyandu activities, BUMDes bookkeeping and community cleanups. Joined mosque and cemetery painting activities.
- Supported school scouting activities and earthquake preparedness education. Helped produce and hand over the village profile video.

PT Angkasa Pura Indonesia Jul 2025 - Aug 2025

Internship Trainee

- Rotated for five weeks in UPS/converter systems and three weeks in electrical protection at Soekarno-Hatta International Airport.
- Visited substations and focused on learning protection relays. Received SCADA instruction and observed PLC installations.

Diponegoro University 2025

Basic Electronics Practicum Assistant

Assisted students with laboratory activities and basic electronic components.

Redesma Technologies Jan 2025 - Feb 2025

Electronics Technician

Worked on IoT temperature monitoring and DC motor control. Co-authored an SOP for comparing Blynk and Thinger.io using ESP32 and a DS18B20 sensor.

SKILLS

Electrical and embedded: Electronics, sensor integration, C, STM32 HAL, STM32CubeIDE, ADC, DMA, timers and UART. Field learning in UPS/converters, protection relays and SCADA.

Software and media: Python, Tkinter, React, FastAPI, Git, automated tests, digital filtering, RMS/MAV extraction, CSV/JSON processing and Canva.

ORGANIZATIONS AND EVENT COMMITTEES

Society of Electronics Engineering Division (SEED) 2025

Head of Media Division

Served as Head of Media for the 2025 student bureau at Diponegoro University.

Google Student Ambassador Program Class of 2025

Google Student Ambassador

Completed the student program and received recognition for championing Google AI within the campus community.

Society of Petroleum Engineers (SPE) Sep 2024 - Dec 2025

Student Member

Joined professional learning activities, including technical writing and geothermal drilling sessions.

ENIGMA Vol. 3, SEED / HME UNDIP Oct 2024

Workshop Instructor

Taught basic electronics workshop activities involving circuits, wiring and PCB work.

Indonesia International Sustainability Forum (ISF) Sep 2024

Organizational Delegate

Participated in the sustainability forum as an organizational delegate.

Himpunan Mahasiswa Elektro, Diponegoro University 2024

Creative Design Staff, Kantor Media Informasi (KMI)

Supported student organization media and design work. Assigned as a co-responsible staff member for the HME Rewind documentation program.

LKMM-PD, HME UNDIP 2023

Event Committee Member

Served on the committee for introductory student management and leadership training.

PMB Teknik Elektro, Diponegoro University 2023

Logistics / Equipment Committee Member

Prepared equipment and facilitated the needs of the new-student welcoming activities.

Decarbonizing Indonesia Nov 2022 - Jan 2023

Campaign Contributor

Created social media campaign and design materials about decarbonization.

PKKMB and Pendikar, Diponegoro University Aug 2022

Participant

Completed the 40-hour new-student orientation and character education program, 15-20 August 2022.

Information and Communication Technology (ICT), SMA Negeri 5 Bekasi Jan 2021 - Dec 2021

Vice Secretary

Prepared work-program and accountability documents, and maintained attendance and member assessment records.

ICT Leadership Regeneration, SMA Negeri 5 Bekasi Nov 2021

Committee Secretary and Session Instructor

Prepared event administration and delivered the secretarial training session for incoming ICT committee members.

CINOTIC, SMA Negeri 5 Bekasi 2021

Event Secretary

Supported proposals, reporting and event administration for the Science Techno & Journalistic Competition.

PROJECTS AND TECHNICAL PROPOSALS

RepairGate Oct 2026

Developed a code-repair verifier with protected tests, independent holdout checks and exported patch decisions. Includes an optional local Ollama adapter and a recorded-candidate demonstration.

ApplyProof Oct 2026

Developed an application workspace with a reviewed fact ledger, explicit skill comparison, office/worksite distinction and traceable application drafts. Optional AI selects fact IDs for verbatim drafting.

DataRescue Oct 2026

Built a local CSV/JSON validation workbench and CLI with rule-based normalization, quarantine of rejected rows and audit-report exports, covered by automated tests.

uart-fixture-kit Oct 2026

Developed reusable synthetic UART fixtures and virtual-time replay to reproduce fragmented, malformed and concatenated byte streams without serial hardware.

Personal Portfolio / Anak Kos Survival 2026

Developed an engineering portfolio with project evidence and a simulated EMG preview. Earlier work includes a seven-day browser game and an interactive portfolio world, retained in the project archive.

Nusantara SolarWing 2026

Prepared a model-based concept package using historical solar data, wind sensitivity and battery reserve planning for maritime patrol scenarios.

EMG Acquisition and Monitoring System Thesis, 2026

Academic thesis, Diponegoro University.

- Implemented 12-bit ADC acquisition with DMA on STM32F411CEU6 and SEN0240 at a nominal 4 kHz sampling rate.
- Applied 20 Hz high-pass, 500 Hz low-pass and 50 Hz notch filtering, with RMS/MAV extraction over 400 samples.
- Built the Python/Tkinter interface for UART telemetry at 115200 baud, live plots, status monitoring and CSV export, with coding-agent assistance. Teammates handled the stimulator and fuzzy control.

PLTA Jatigede Case Study 2025

Contributed the introduction and task allocation to a group presentation covering NPV, benefit-cost ratio and payback period.

Redesma Technologies / Timor-Leste Concept 2025

Authored a proposal for rural technology options, including solar-powered irrigation, community warning systems and digital monitoring.

EnvironmentID: IoT-powered Biofiltration 2024

Co-authored an abstract proposing smart sensors and oyster ecosystems for coastal water-quality monitoring and biofiltration.

GOVTECH / Gov-AI Hackathon Proposal 2024

Prepared a proposal for integrated digital public services using APIs, responsive interfaces and AI-assisted information services.

COURSES, WORKSHOPS AND CERTIFICATIONS

Sep 2026: **Introduction to Internet of Things**, Cisco Networking Academy. Completed 27 September 2026.

Sep 2026: **AI Fundamentals**, Google via Coursera. Completed 26 September 2026.

Sep 2025: **SPENTRODUCE: Exploring the Downstream Oil and Gas Industry**, SPE UNDIP Student Chapter. Participant.

Jan 2025: **AI-Powered Shopping Ads Certification**, Google Skillshop. Historical credential, expired January 2026.

Oct 2024: **Workover and Well Service Talks**, IATMI Student Chapter ITB. Participant.

Sep 2024: **Automation Clinic: Motor Protection and Control**, Engineering Academy. Participant.

Sep 2024: **How to Write a Good Technical Paper**, SPE. 1.5-hour continuing education session.

Sep 2024: **Geothermal Drilling in Volcanic and Metamorphic Basins**, SPE. 1.5-hour continuing education session.

Sep 2024: **Strategies in Geothermal Exploration: Maximizing Indonesia's Potential**, SPE UNDIP Student Chapter. Guest lecture participant.

Sep 2024: **Ethics of Artificial Intelligence: Navigating Responsible Technology for the Future**, SkilHub Community by Skilvul / Nocturnailed. Webinar participant.

May 2024: **Fundamentals of Digital Marketing**, Google Digital Academy / Skillshop. Completed 27 May 2024.

May 2024: **Introduction to Electronics**, The Open University OpenLearn. Completed a 10-hour course on 21 May 2024.

May 2024: **IT: Device to Device Communication**, The Open University OpenLearn. Completed a 20-hour course on 19 May 2024.

Mar 2024 - May 2024: **College Copilot Bootcamp (CABOOT)**. Completed six sessions, 20 March - 4 May 2024.

Nov 2020 - Feb 2021: **KSN-K Geography Preparation**. Participant in the 8 November 2020 - 7 February 2021 training program.

Date not recorded: **ML/AI Fundamentals: How Machine Learning Is Shaping the Oil & Gas Industry**, SPE Java Section.

ADDITIONAL RECOGNITION

May 2024: **100 Design Achievement Badge**, Canva.

2019: **Try Out Akbar Winner**, Penerbit Erlangga. Score: 100, SMP/MTs category, Jakarta. SMP Angkasa Halim Perdanakusuma.